18 Harmonic oscillators

(Pre-requisite: None really, but Classical Mechanics would make things easier)

A point P moves so that its position vector $\mathbf{r}$ relative to the origin O satisfies the equation $\ddot{\mathbf{r}} = -\omega^2\mathbf{r}$, where ω is a constant. Show that the motion of P is compounded from two harmonic oscillations executed with the same frequency at right angles to each other and that the locus of P is an ellipse with the centre at O.

Show further that if P has Cartesian coordinates (x, y) in this plane and O is the origin of coordinates, the locus of P has the equation

$$\frac{x^2}{a_1^2} + \frac{y^2}{a_2^2} - \frac{2xy}{a_1 a_2}\cos(\epsilon_2 - \epsilon_1) = \sin^2(\epsilon_2 - \epsilon_1),$$

where a_1 and a_2 are the amplitudes of the harmonic oscillations and ϵ_1 and ϵ_2 are their respective phases.

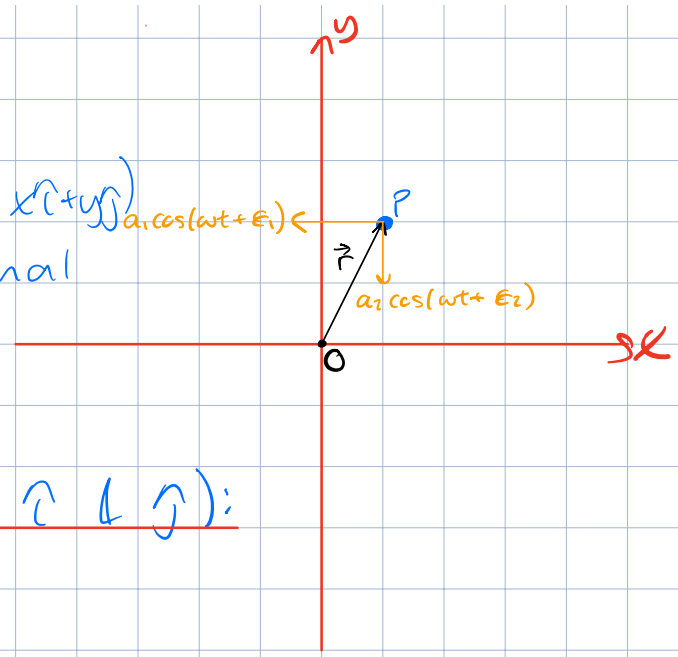
Discuss the shape and orientation of this locus for different values of $\epsilon_2 - \epsilon_1$.

If the two harmonic oscillations are now taken to have different frequencies, obtain the equation of the path of P.

$\ddot{\vec{r}} = \ddot{x}\hat{i} + \ddot{y}\hat{j}$

$\ddot{x}\hat{i} + \ddot{y}\hat{j} = -\omega^2(x\hat{i} + y\hat{j})$

$\hat{i}$ & $\hat{j}$ orthogonal



So equating $\hat{i}$ & $\hat{j}$):

$\ddot{x} = -\omega^2 x$

$\ddot{y} = -\omega^2 y$

$x = a_1\cos(\omega t + \epsilon_1)$

$y = a_2\cos(\omega t + \epsilon_2)$

$a_1, a_2, \epsilon_1, \epsilon_2$ depend on initial conditions.

Derivation of solution ($\ddot{\vec{r}} = -\omega^2 \vec{r}$) $\vec{r} = x\hat{i} + y\hat{j}$

Since the 2 oscillators act orthogonally:
We can seperate x and y components

x component

$\ddot{x} = -\omega^2 x$

2nd order homogeneous, linear ODE

$x(t) = a_1 \cos(\omega t + \varepsilon_1)$

Likewise with the y component:

$y(t) = a_2 \cos(\omega t + \varepsilon_2)$

Let $\theta = \omega t + \varepsilon_1$

$\delta = \varepsilon_2 - \varepsilon_1$

$x = a_1 \cos(\theta)$

$\Rightarrow \frac{x}{a_1} = \cos(\theta)$ (1)

$y = a_2 \cos(\theta + \delta)$

$\Rightarrow \frac{y}{a_2} = \cos(\theta + \delta)$

$= \cos\theta \cos\delta - \sin\theta \sin\delta$

$= \frac{x}{a_1} \cos\delta - \sin\theta \sin\delta$ (2)

$\sin^2\theta = 1 - \cos^2\theta$

$= 1 - \frac{x^2}{a_1^2}$

$\sin\theta = \pm\sqrt{1 - \frac{x^2}{a_1^2}}$

Taking $+\sqrt{\ }$ gives

$$\sin\theta = \sqrt{1-\frac{x^2}{a_1^2}}$$

So, subbing into (2) gives:

$$\frac{y}{a_2} = \frac{x}{a_1}\cos\delta - \sqrt{1-\frac{x^2}{a_1^2}}\sin\delta$$

$$\frac{x}{a_1}\cos\delta - \frac{y}{a_2} = \sqrt{1-\frac{x^2}{a_1^2}}\sin\delta$$

= Squaring both sides:

$$\frac{x^2}{a_1^2}\cos^2\delta - \frac{2xy}{a_1a_2}\cos\delta + \frac{y^2}{a_2^2} = \left(1-\frac{x^2}{a_1^2}\right)\sin^2\delta$$

$$\frac{x^2}{a_1^2}\cos^2\delta - \frac{2xy}{a_1a_2}\cos\delta + \frac{y^2}{a_2^2} = \sin^2\delta - \frac{x^2}{a_1^2}\sin^2\delta$$

$$\frac{x^2}{a_1^2}\left(\cos^2\delta + \sin^2\delta\right) - \frac{2xy}{a_1a_2}\cos\delta + \frac{y^2}{a_2^2} = \sin^2\delta$$

$$\frac{x^2}{a_1^2} + \frac{y^2}{a_2^2} - \frac{2xy}{a_1a_2}\cos\delta = \sin^2\delta$$

$$\frac{x^2}{a_1^2} + \frac{y^2}{a_2^2} - \frac{2xy}{a_1a_2}\cos(\varepsilon_2-\varepsilon_1) = \sin^2(\varepsilon_2-\varepsilon_1)$$

$\delta = 0 \iff \varepsilon_2 = \varepsilon_1$

We have $\frac{x^2}{a_1^2} + \frac{y^2}{a_2^2} - \frac{2xy}{a_1 a_2} = 0$

$$\left(\frac{x}{a_1} - \frac{y}{a_2}\right)^2 = 0$$

$$\frac{x}{a_1} - \frac{y}{a_2} = 0$$

$$y = \frac{a_2}{a_1} x$$

Elipses degenerates
Domain would depend on initial conditions.

$\delta = \frac{\pi}{2} \iff \varepsilon_2 - \varepsilon_1 = \frac{\pi}{2}$

We have $\frac{x^2}{a_1^2} + \frac{y^2}{a_2^2} = 1$
Ellipses in standard form.

Solution when the frequencies differ

$$\left.\begin{aligned}\ddot{x} &= -\omega_x^2 x \\ \ddot{y} &= -\omega_y^2 y\end{aligned}\right\} \omega_x \neq \omega_y$$

Lissajous curves

$$\Rightarrow \left.\begin{aligned} x &= a_1 \cos(\omega_x t + \varepsilon_1) \\ y &= a_2 \cos(\omega_y t + \varepsilon_2)\end{aligned}\right\} \text{due to orthogonality}$$

If $\frac{\omega_x}{\omega_y}$ is irrational, the path never repeats. The point P will eventually pass arbitrarily close to every point within the box $[-a_1, a_1] \times [-a_2, a_2]$

When $\frac{\omega_x}{\omega_y}$ is rational the path repeats

For which values of ω_x & ω_y does P pass through the origin?

Extension ideas

- What happens when there are frictional forces acting
- What happens when the driving forces are non-orthogonal
- Link to energy / Hamiltonian
- Specific example e.g magnetic field
- Intersection points of loci for different values of ω_x & ω_y. → how many are there & do they form a shape?

  e.g. for $P_1: \omega_x = 1, \omega_y = 2$

  $P_2: \omega_x = 1, \omega_y = 3$

  There are 9 intersections